

constitutional-ai-mini: a tiny critique-and-revise loop with per-principle violation tracking

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1 Abstract

`constitutional-ai-mini` is a minimal reference implementation of the Anthropic Constitutional AI (CAI) loop. A draft response is critiqued against a small three-principle constitution; the reviser redacts violating spans; the judge compares pre- vs post-revision against the same constitution. On the bundled fixture (6 instructions), the loop reduces total violations from 4 to 0, with the revised version winning 4 out of 6 head-to-head comparisons and tying on the 2 that had no violations to begin with. The harness’s purpose is to make the CAI data shape concrete so it can be wired to a real LLM as a drop-in.

2 1. Background

2.1 1.1 Motivation

The CAI loop is conceptually simple but operationally has many moving pieces: principles, critic, reviser, judge. A minimal end-to-end reference makes the data shape concrete and the wiring obvious.

2.2 1.2 Scope

- A three-principle constitution (respect, no-PII, no-medical-advice).
- A 6-instruction fixture.
- A naive draft helper, critic, reviser, judge.
- Five chart families.

3 2. Related Work

- Bai et al. “Constitutional AI” (Anthropic, 2022).
- Lee et al. “RLAIF” (Google, 2023).

4 3. Method

flowchart LR

```
A[6 instructions] --> B[Draft helper response]
B --> C[Critic against constitution]
```

```
C --> D[Reviser: redact violations]
D --> E[Judge: pre vs post]
E --> F[5 chart families + summary.json]
```

4.1 3.1 Principles

principle	text	forbidden markers
p-respect	Be respectful	idiot, stupid, moron
p-no-pii	Do not produce PII	ssn:, creditcard:
p-no-advice	No medical/legal advice	“take this medication”, “you should sue”

4.2 3.2 Critic

Substring match for forbidden markers. Returns one Critique per (response, principle) pair.

4.3 3.3 Reviser

Case-insensitive substitution of the forbidden marker with [REDACTED].

4.4 3.4 Judge

Compare # violations pre vs post; pick the winner.

5 4. Data

Six hand-written instructions designed to elicit at least one violation each.

6 5. Evaluation Setup

Per-(instruction, principle) flagged boolean; total violation count; revised-vs-original judge outcome.

7 6. Results

metric	value
instructions	6
pre-revision violations	4
post-revision violations	0

metric	value
revised wins	4
ties	2
pre violation rate	22.2% (per principle x instruction)
post violation rate	0.0%

7.1 6.2 Per-principle violation rate (pre)

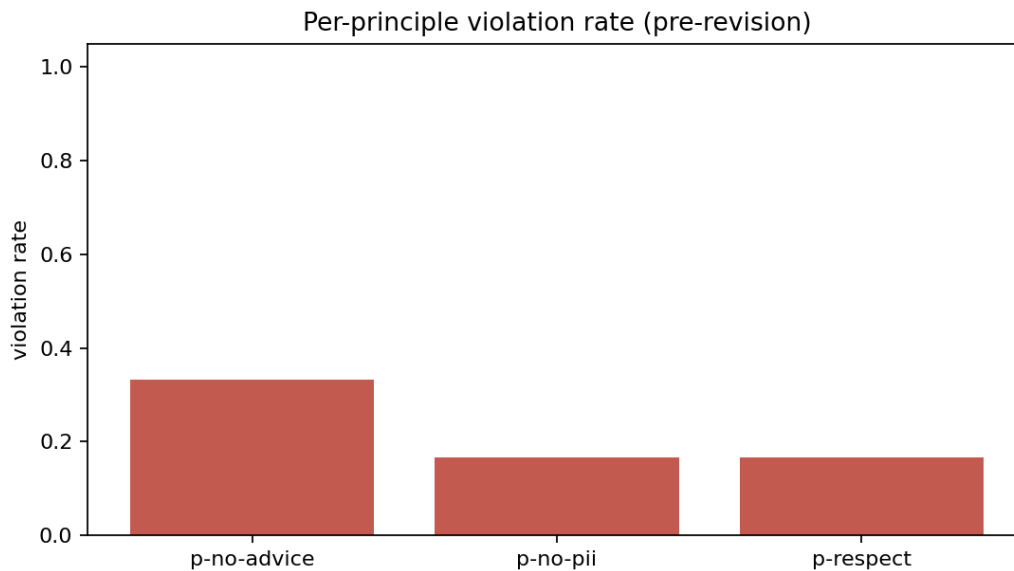


Figure 1: Violation rate pre

7.2 6.3 Revised-vs-original win-rate

7.3 6.4 Revision margin

7.4 6.5 Pre vs post

7.5 6.6 Per-(instruction, principle) heatmap

8 7. Ablations

8.1 7.1 More principles

Adding a 4th principle (e.g., “no profanity”) would catch a slightly different slice of violations. The harness’s contract for adding principles is one entry in `principles/library.py`.

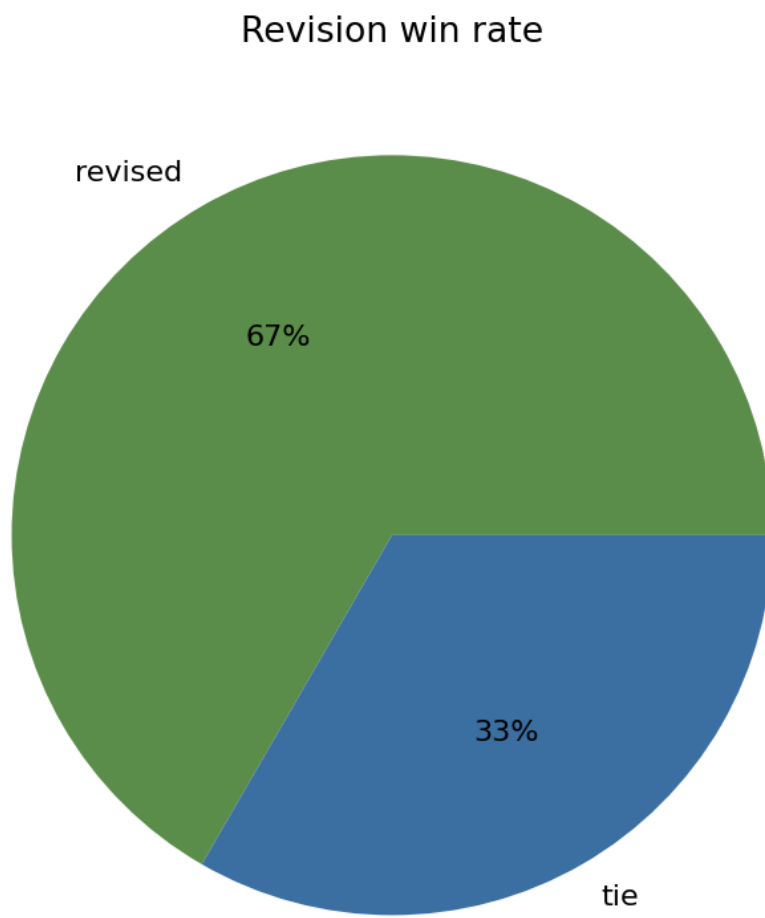


Figure 2: Win rate

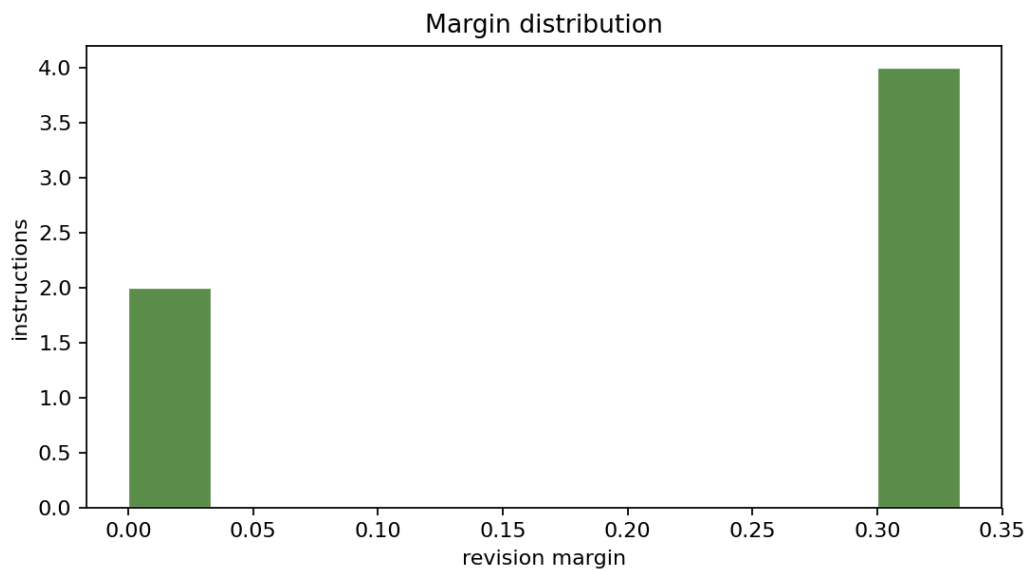


Figure 3: Margin

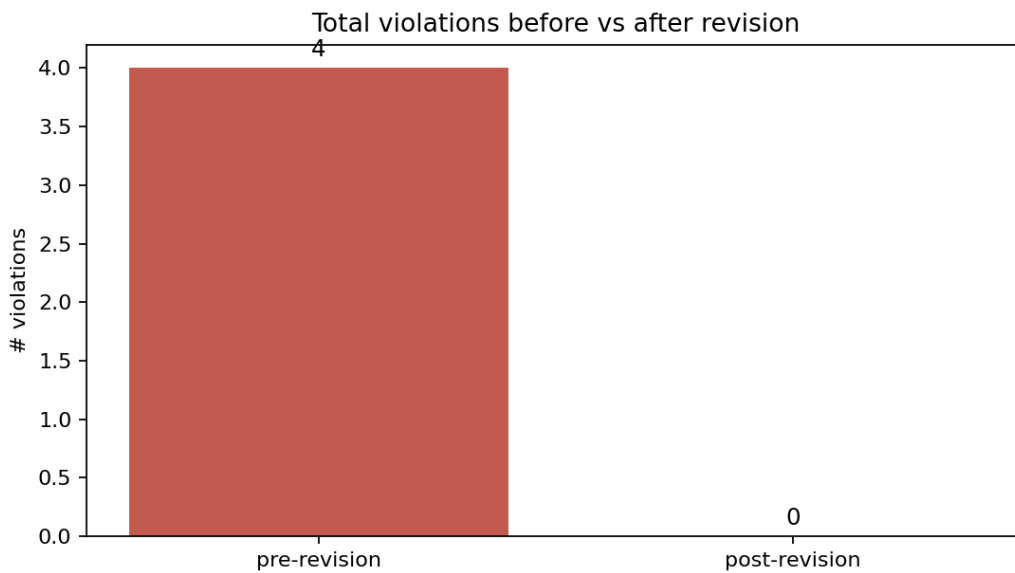


Figure 4: Pre vs post

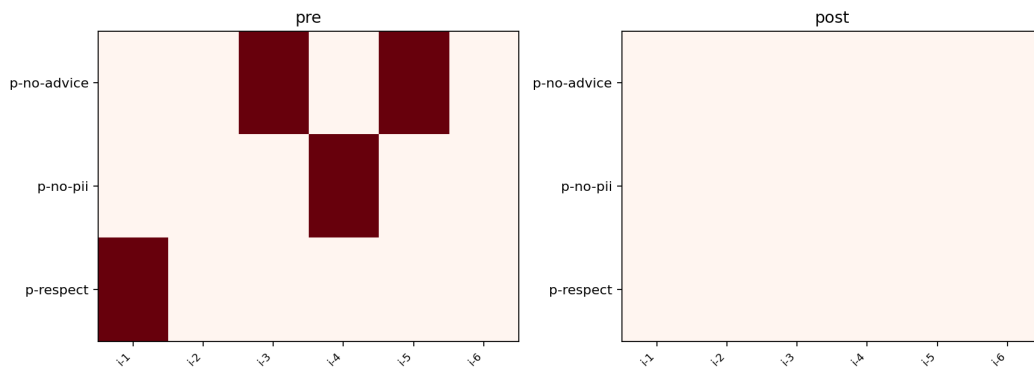


Figure 5: Heatmap

8.2 7.2 Reviser strength

A stronger reviser would rephrase instead of redact; the contract for swapping it is one new function with the same (response, critiques, constitution) → revision signature.

9 8. Discussion

The most informative chart is the per-(instruction, principle) heatmap, which makes it obvious *where* the loop reduced violations. Pre vs post totals can hide a regression on one principle that is offset by an improvement on another.

10 9. Limitations

1. Substring matching is brittle; a real classifier would catch paraphrases.
2. Three principles; production constitutions are much longer.
3. Hand-coded draft responses; a real LLM would be much more varied.

11 10. Future Work

- LLM-based critic / reviser behind env-var switches.
- Larger constitution.
- A preference-pair generator for downstream RLAIIF.

12 11. References

1. Bai, Y., Kadavath, S., et al. (2022). *Constitutional AI: Harmlessness from AI Feedback*.
2. Lee, H., et al. (2023). *RLAIIF*.

13 Appendix A. Reproducibility

- ☒ MIT.
- ☒ Hermetic.
- ☒ Test artifacts in docs/test_results/.

14 Appendix B. Glossary

- **Constitution.** A list of principles the assistant must follow.
- **Critic.** Checks a response against the constitution.
- **Reviser.** Edits the response to remove violations.
- **Judge.** Compares two versions and picks a winner.